Walking Talk
Group 10 · co-chairs Sarah Blossom and Bethany Jorgensen

13 October 2026
12:30–14:00

Screening commercial smoked seafood for microplastics: a multi-taxa market survey

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Sold sealed, ready to eat

45 packs, 15 products, Istanbul retail

Eaten as sold, often whole

what the pack carries, the consumer eats

What else is in the pack?

KOH digestion, alumina filter, stereo microscope

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Walking Talk group 10 on 13, 14 and 15 October, 12:30–14:00. Write to me with questions or your own retail-pack numbers.

Funding. This study was funded by Scientific Research Projects Coordination Unit of Istanbul University. Project number: FYL-2025-41504. **Conflict of interest.** The authors have no conflict of interest to disclose related to this presentation.

Why smoked seafood?

- **Eaten as sold.** Ready to eat, often whole or near-whole. What the pack carries, the consumer eats.
- **A gap in the data.** Processed seafood is under-represented in microplastic surveys next to fresh fish and shellfish.
- **More contact surfaces.** Brining, smoking racks, slicing and vacuum film: contact points fresh-product surveys never sample.
- **Aim.** A transparent, reproducible baseline for ready-to-eat seafood on the Turkish market, as the basis for dietary-exposure assessment.

What we sampled

45 packs · 15 categories · every category in triplicate

Bought September 2025 to April 2026 from chain retailers and specialist fishmongers in Istanbul; carried in a cool box, kept sealed at −20 °C; species, brand, producer, lot and pack weight recorded. Atlantic salmon from five anonymised suppliers (A to E) plus supplier A's ASC-certified line, so sourcing and certification can be compared within one species.

Atlantic salmon 6 categories · **Other finfish** 7 categories · **Crustacean** 1 category · **Cephalopod** 1 category

Salmon A *Salmo salar* 100 g pack	**Salmon B** *Salmo salar* 200 g pack	**Salmon C** *Salmo salar* 100 g pack	**Salmon D** *Salmo salar* 100 g pack	**Salmon E** *Salmo salar* 100 g pack
Salmon A ASC *Salmo salar* 100 g pack	**Rainbow trout** *Oncorhynchus mykiss* 100 g pack	**Black Sea trout** *Salmo labrax* 100 g pack	**Mackerel** *Scomber scombrus* 200 g pack	**Mackerel (çiroz)** *Scomber scombrus* 1000 g pack
Tuna (pouch) *Thunnus alalunga* 80 g pack	**Sea bream** *Sparus aurata* 100 g pack	**Sea bass** *Dicentrarchus labrax* 100 g pack	**Shrimp** *Penaeus spp.* 100 g pack	**Octopus** *Octopus vulgaris* 250 g pack

How we looked

10 g from each pack per digest, contamination-controlled, pictured at our bench

1 Open and weigh

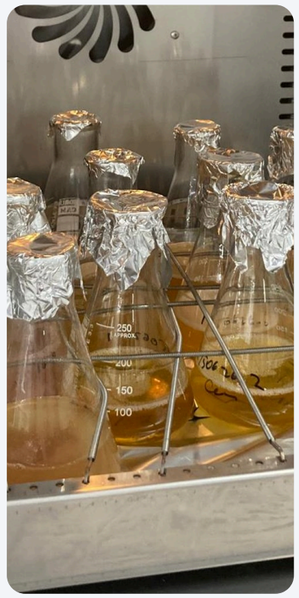
Thawed overnight at +4 °C. Pack interior rinsed three times with filtered ultrapure water, rinse pooled with the sample. 10 g weighed to 0.01 g per digest.



A pack as bought: tray, film and fillets.

2 Alkaline digestion

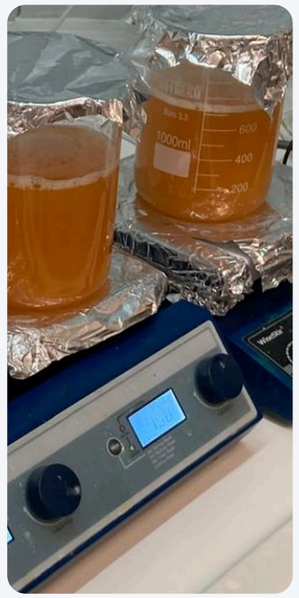
10 % KOH with 2 % Tween-20, 1:5 (w/v), 40 °C, 150 rpm, 24 h in foil-capped glass flasks. The surfactant keeps the fat emulsified so the filter does not clog.



Foil-capped flasks in the shaking incubator.

3 Neutralise

Diluted 1:1 with filtered water, then 1 M citric acid under stirring to pH 7.5, the stability window of the alumina membrane.



Digests on the multi-stirrer, pH meter alongside.

4 Filter and clean

0.2 µm Al_2O_3 membrane, 47 mm, at −500 mmHg; funnel rinsed 3 × 10 mL. 15 % H_2O_2 on the filter for 30 min, then dried 24 h covered.



Filtration onto the 47 mm alumina membrane.

5 Count and classify

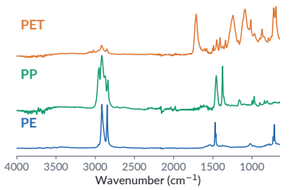
Stereo microscope: whole filter at 10×, each particle at 40×. Morphotype, colour and maximum length (ImageJ) recorded per particle.



Filter surface at 10×: fibres and a blue fragment.

6 Next: polymer identity

µ-FTIR and ATR on 3–5 particles per filter, stratified by size; then exposure per 100 g and 225 g portion in the thesis.



Reference ATR-FTIR spectra from our polymer library.

What we found, in five numbers

first-pass counts, 309 suspected particles

45

retail packs, 15 smoked products in triplicate

100 %

of packs with suspected particles (blanks pending)

50

suspected particles per 100 g, median pack

81 %

Fibres, the dominant morphotype

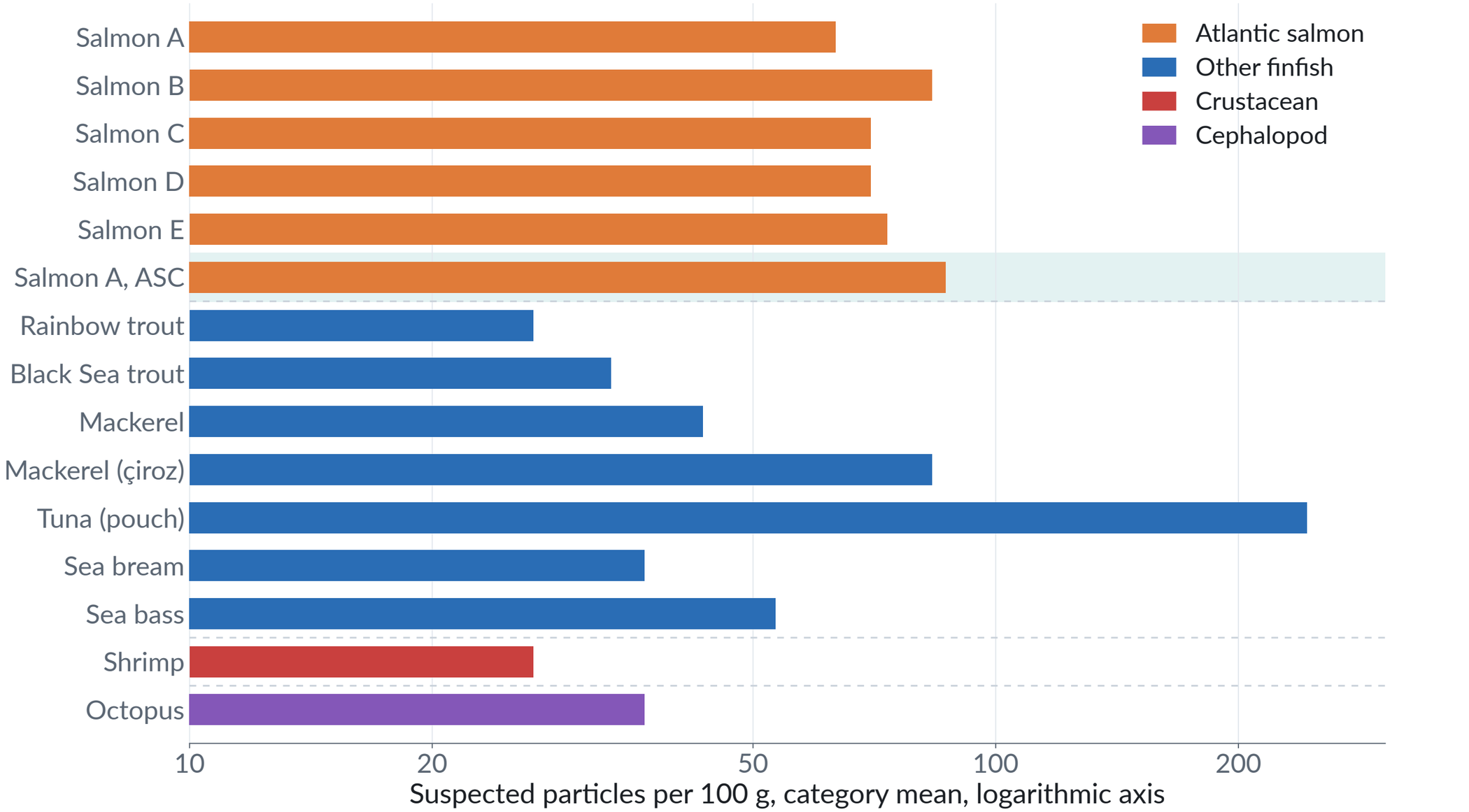
72 %

of particles 1–5 mm long

Read these as an upper bound. Raw counts under the stereo microscope; blank subtraction is pending and visual identification over-counts. Polymer identity by µ-FTIR is the next step, and only confirmed particles will enter the exposure estimate.

What we found, by product

suspected particles per 100 g, category mean · preliminary

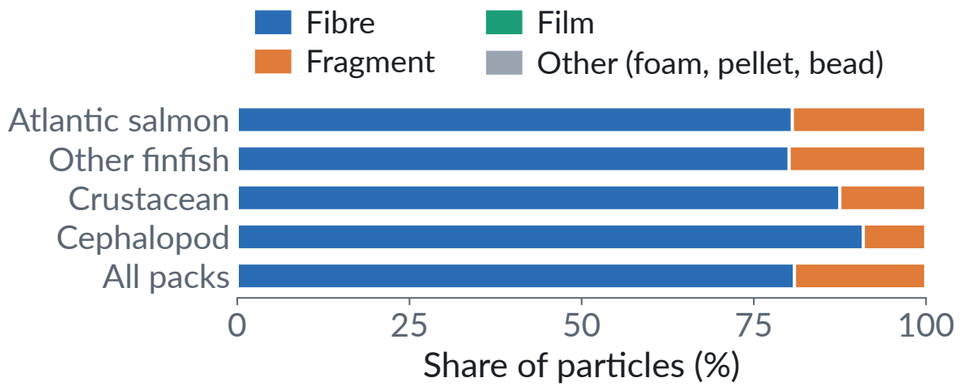


Bars: mean of the three packs of a category (raw counts, blank subtraction pending), on a logarithmic axis. Shaded row: the ASC-certified line, from the same supplier as Salmon A. No individual pack readings are shown on purpose; the thesis carries them.

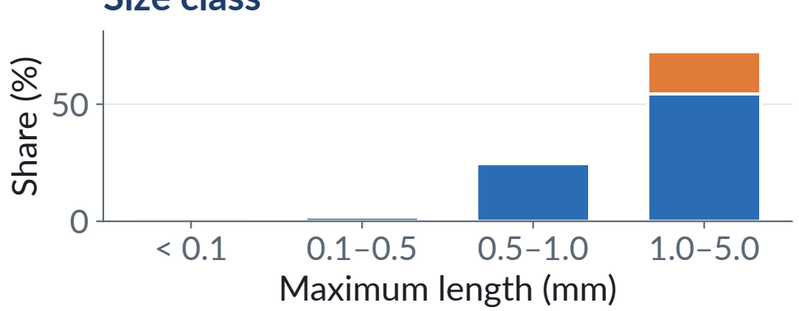
What they look like

309 particles on 45 filters, uncorrected

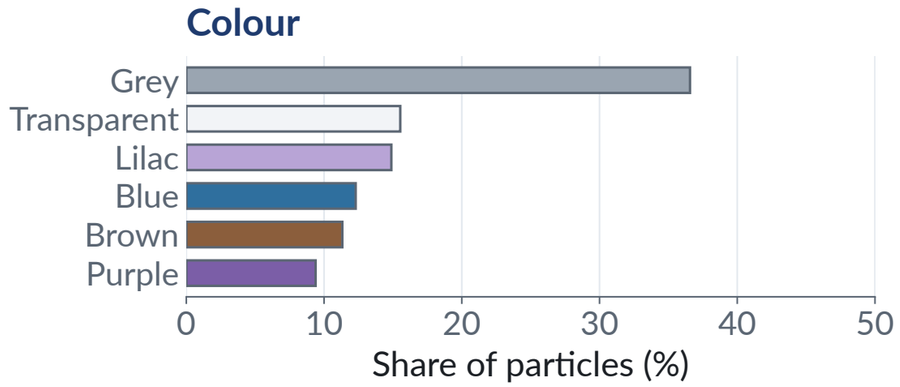
Morphotype, by taxon group



Size class



Colour



Take-home

1. **Occurrence.** Suspected microplastics were recovered from 100 % of the 45 packs analysed, in every taxon group; fibres made up 81 % of all particles, 72 % were 1–5 mm long and grey was the most frequent colour (37 %).
2. **Within Atlantic salmon.** Supplier means ranged from 63 to 87 particles per 100 g across suppliers A to E; the ASC-certified line (87) was not lower than the same supplier's standard line (63).
3. **Preliminary, and an upper bound.** First-pass counts; blank correction is pending. Visual identification over-counts: polymer identity by µ-FTIR comes next, and only confirmed particles enter the per-portion exposure estimate (100 g and 225 g) in the thesis.

Talk to me. Cem Hasşerbetçi · **cemhasserbetci@istanbul.edu.tr** · group 10, all three days.
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Funded by Scientific Research Projects Coordination Unit of Istanbul University, project FYL-2025-41504. No conflict of interest to disclose.